

# The Analysis on the Collective Forest Property Rights Institution Reform with Game Theory

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**Abstract**-The forestry development and the environment improvement are bound by a common cause. When the demand of improving the environment increases, the people pay more attention to the forestry development. In this text, I analyzed the reasons, content and performance of the reform of collective forest property rights institution with game theory, and concluded that devolving the using rights of forest land and property rights of timber trees to farmers though implementing reform of collective forest property rights institution would increase the production enthusiasm of farmers, and it is helpful to promote the coordinated development of forestry and improve the environment.

**Keywords**-Reform of Collective Forest Property Rights Institution; Game Theory; Public Goods

At different stages of economic development, different types of forest property rights reflect different efficiency. It should meet the demand for forestry development to select the suitable type of forest property rights which is compatible with the economic development. Mckan, Uish Maskey, TesfaG. Gebremedhin, TimothyJ. Dalton, Peter Gluck<sup>[1]</sup> and others analyzed the characteristics and advantages as well as disadvantages of state-owned forest property rights, collective-owned forest property rights and private forest property rights deeply; Jiang Hai, Hong Zhisheng<sup>[2]</sup>, Liu Jian-min<sup>[3]</sup>, Gushan Song<sup>[4]</sup> and others discussed the business model of different types of forest property rights in China. With the level of economic development was rising, the institutional environment of forest property rights changed, and then forest property rights would change to be the more efficient type of forest property rights in order to meet the demand of the people to improve ecological conditions. In order to mobilize the farmer's enthusiasm to operate forest, reforestation, forest protection, and increase the farmer's income as well as improve the ecological conditions, we carried out the collective forest property rights institution reformation which making the conversion of forest land use rights and timber property rights break down into each peasant and farmers in the same collective economic organizations as core in 2003 in Fujian, Jiangxi, Zhejiang and other provinces of China.

## I. THE ANALYSIS ON DYNAMICS OF COLLECTIVE FOREST PROPERTY RIGHTS INSTITUTION REFORM WITH GAME THEORY

Before the collective forest property rights institution reform, we adopted the business model of unified management

by the collective economic organizations for collective forest in most area in China. At that time, the relationship between collective organization and workers who engaged in the production and management of forestry is the principal - agent relationship. The situation meets the following basic assumptions: (1) the cost when workers are diligent is greater than the cost when workers are lazy, and the income of the collective organization when workers are diligent is also greater than the income when workers are lazy; (2) the workers receive a fixed wage income; (3) when the collective organization supervise the workers, if they discover the workers are lazy, they should punish them; (4) the net income for the collective organization equals the income which workers earned for the collective organization minus the wages which be paid to workers, and then minus the monitoring cost when the collective organization carrying out monitoring and supervision to the workers, and then plus the fines which the workers paid to the collective organization when they were discovered that they were lazy; (5) the net income for the workers equals the wages minus the work cost, and then minus fines when they were discovered that they were lazy. If the workers work hard, the collective organization gets the benefits A, the work cost of the workers is a; if the workers are lazy, then the collective organization gets the benefits B, the work cost of the workers is b; and  $A > B$ , while  $a > b$ . The workers receive a fixed wage I; the fine which the workers should pay to the collective organization when they were supervised that they were lazy is e; the monitoring cost of the collective organization is c. The game process between the collective organization and the workers is shown in Figure 1.

|                         |                     | Workers    |                |
|-------------------------|---------------------|------------|----------------|
|                         |                     | Diligent   | Lazy           |
| Collective Organization | Supervise           | A-I-c, I-a | B-I-c+e, I-b-e |
|                         | Without Supervision | A-I, I-a   | B-I, I-b       |

**Figure1 The analysis with game theory between collective origination and workers**

From the perspective of the workers, when the collective organization surprised, the net income for the workers was (I-a) when they were diligent, and the net income for the workers was (I-b-e) when they were lazy, and then the worker's

decision depended on the comparison between  $a$  and  $(b + e)$ . When  $a > (b + e)$ , the workers should choose to be lazy; When  $a < (b + e)$ , the worker should choose to be diligent. So if you want to motivate the workers to work hard, you must increase the penalties when they were lazy, it is  $e > (a - b)$ . When the collective organization did not surprise, just because  $a > b$ ,  $(I - a) < (I - b)$ , the best choice for the workers is to be lazy. From the perspective of the collective organization, when the workers were diligent, due to  $(A - I) > (A - I - c)$ , therefore the collective organization did not surprise; when the workers were lazy, if the collective organization choose to surprise or not depends on the relationship between the fines which requested to the workers when they were discovered that they were lazy and the monitoring cost, that is the relationship between  $e$  and  $c$ , when  $e > c$ , the collective organization choose to supervise; when  $e < c$ , the collective organization choose not to supervise.

It can clearly be seen that under the collective forest property rights institution arrangement, the business model of unified management by the collective organization calls for that the collective organization should surprise the workers objectively, impartially and accurately in order to prompt the workers to work hard. For the collective organization, it is relatively easy to surprise the work time, but it is very difficult to surprise the work quality. In this case, “to work but not work hard” becomes a rational choice for the workers. Thus, the business model of unified management by the collective organization is not helpful for mobilizing the enthusiasm of the farmers [5]. It is urgent to explore a more efficient forest property rights institutional arrangement to replace the business model of unified management by the collective organization.

## II. THE ANALYSIS ON CONTENT OF COLLECTIVE FOREST PROPERTY RIGHTS INSTITUTION REFORM WITH GAME THEORY

Delegating forest land use rights and timber property rights to the farmers is helpful for internalizing the external economy occurred when the farmers select “work but not work hard”, and also helpful for mobilizing the farmer's enthusiasm for production. But it does not mean that it must delegate all of the forest land use rights and timber property rights to the farmers. Ecological forests are the forests which play the main ecological benefits of forests [6], thus, ecological forests should be supplied by the government [7], and it can also be analyzed with the prisoner's dilemma model [8].

The basic assumes of the model as following: (1) the participants were private; (2) the economic benefits of the ecological forests were little, so it can not be considered; (3) the supply of the ecological forest requires pay the costs; (4) just because the environmental product have the characters of indivisibility, the utility which participants obtained from the environmental product is the same, whatever the participants supply or not supply the ecological forest; (5) When all the private do not supply the ecological forests, the ecological forests should be supplied by the government which outside the model.

Assumed that the maximum utility which the participants could obtained when they did not supply the ecological forests

is 2; if they supplied the ecological forests, the cost they should pay is 1, so the utility they obtained ultimately is 1. This moment, the game process between participants A and B is shown in Figure 2.

|               |            | Participant A |            |
|---------------|------------|---------------|------------|
|               |            | Supply        | Not Supply |
| Participant B | Supply     | 1, 1          | 1, 2       |
|               | Not Supply | 2, 1          | 2, 2       |

**Figure2 The analysis with game theory between participants**

From the perspective of the participant A, when participant B supply the ecological forests, the best choice for the participant A is “not supply”; while when participant B do not supply the ecological forests, the best choice for the participant A is still “not supply”. Participant B faces the same situation, so, in this model there is a dominant equilibrium of (2, 2), it is that the rational choice for private is “not supply”, while the ecological forest is supplied by the government. It is the reason why the content of the collective forest property rights institution reform do not include the ecological forest which play the main ecological benefits of forests.

## III. THE ANALYSIS ON PERFORMANCE OF COLLECTIVE FOREST PROPERTY RIGHTS INSTITUTION REFORM WITH GAME THEORY

Through the collective forest property rights institution reform, we delegated forest land use rights and timber property rights which still unified management by the collective organizations to the farmer household by the way of family contract. It changed the payoff matrix of the game model under the business model of unified management by the collective organizations, and the situation is shown in Figure 3.

The mean of each letter is basically the same with the model of the business model of unified management by the collective organizations (Figure 1). It is assumed that the collective organization delegate forest land use rights and timber property rights to the farmer household by the way of family contract and obtaining the revenue-sharing. The proportion which farmer household occupied is  $\beta$  ( $0 \leq \beta \leq 1$ ), and the proportion which collective organization occupied is  $(1 - \beta)$ . There should be an additional assumption is that when the farmer household did not obtain the same income from the collective organization, the net income when the farmers were diligent is greater than the net income when the farmers were lazy, it is  $A - a > B - b$ .

|                         |                     | Farmers                            |                                            |
|-------------------------|---------------------|------------------------------------|--------------------------------------------|
|                         |                     | Diligent                           | Lazy                                       |
| Collective Organization | Supervise           | A $(1 - \beta) - c$ , $A\beta - a$ | B $(1 - \beta) - c + e$ , $B\beta - b - e$ |
|                         | Without Supervision | A $(1 - \beta)$ , $A\beta - a$     | B $(1 - \beta)$ , $B\beta - b$             |

**Figure3 The analysis with game theory between collective origination and farmers after implementing the reform of collective forest property rights institution**

From the perspective of the farmers, when the collective organization supervised the farmers, whether farmers choose to be diligent or to be lazy depend on the comparison between  $(A\beta - a)$  and  $(B\beta - b - e)$ . When  $(A\beta - a) > (B\beta - b - e)$ , that is,  $\beta > (a - b - e) / (A - B)$ , the farmers should choose to be diligent; when  $\beta < (a - b - e) / (A - B)$ , the farmers should choose to be lazy. When the collective organization did not supervise the farmers, whether farmers choose to be diligent or to be lazy depend on the comparison between  $(A\beta - a)$  and  $(B\beta - b)$ . When  $(A\beta - a) > (B\beta - b)$ , that is,  $\beta > (a - b) / (A - B)$ , the farmers should choose to be diligent; when  $\beta < (a - b) / (A - B)$ , the farmers should choose to be lazy. It is different from the business model of unified management by the collective organization even in the case of the collective organizations did not surprise, as long as the farmers would be given the right to occupy the enough proportion in the distribution of income, they would like to be diligent. From the perspective of the collective organization, when farmers were diligent, just because of  $A(1 - \beta) > A(1 - \beta) - c$ , the best choice for the collective organization is “without supervision”; when farmers were lazy, whether the collective organization choose to surprise or not still depend on the comparison between  $e$  and  $c$ , when  $e > c$ , the collective organization should choose to supervise; when  $e < c$ , the collective organization should choose “without supervision”.

From the perspective of the collective organization and the farmers, to give the farmers the right to occupy the enough proportion in the distribution of income is helpful to mobilize the farmer's enthusiasm. When the farmers were diligent, the best choice for the collective organization is “without supervision”. In the extreme cases, when  $\beta = 1$ , the payoff matrix is shown in Figure 4.

|                         |                     | Farmers  |             |
|-------------------------|---------------------|----------|-------------|
|                         |                     | Diligent | Lazy        |
| Collective Organization | Supervise           | -c, A-a  | -c+e, B-b-e |
|                         | Without Supervision | 0, A-a   | 0, B-b      |

**Figure4 The analysis with game theory between collective origination and farmers under the condition of  $\beta=1$**

When the collective organization supervised, due to  $A-a > B-b$ , then  $A-a > B-b-e$ , the best choice for the farmers is to be diligent; when the collective organization did not supervise, because  $A-a > B-b$ , the best choice for the farmers is still to be

diligent. In this way, “be diligent” becomes a dominant balance for the farmers, at this time, “without supervision” is the best choice for the collective organization, and ultimately become the balance of  $(0, A-a)$ .

In the case of delegating forest land use rights and timber property rights to the farmer household, as long as the farmers were given the right to occupy the enough proportion in the distribution of income, their enthusiasm would greatly enhance; even in some areas, in order to mobilize the farmer's enthusiasm in the greater extent, the collective organization delegated the forest land use rights to farmer household for free.

#### IV. CONCLUSIONS

After we analyzed the dynamics, content and performance of the collective forest property right institution reform with game theory, we found that it is helpful for mobilizing the farmer's enthusiasm to operate forest, reforestation, forest protection, and increasing the farmer's income as well as promoting forestry coordinated development to carry out the collective forest property right institution reform. The collective forest property right institution reform has made a great contribution to improve the ecological environment in large part.

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